

Training-Free Temporal and Semantic Enhancements for Open-Weight Video Diffusion Models

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Abstract—Recent advancements in video generation have been driven by large-scale open-weight Diffusion Transformers (DiTs). However, these models frequently suffer from temporal instability, geometric distortion, and weak semantic adherence to complex prompts. In this paper, we propose a zero-cost, training-free methodology to enhance the temporal consistency and semantic alignment of open-weight video diffusion models, specifically evaluating our approach on the 10-billion parameter Mochi-1-preview architecture. Our method isolates and addresses these artifacts through three independent inference-time modifications: (1) shifting the Flow Match Euler Discrete Scheduler distribution, (2) injecting structured asymmetric negative text embeddings, and (3) amplifying classifier-free guidance. Through a rigorous ablation study across diverse prompt topologies and random seeds, we demonstrate that our combined approach improves structural similarity (SSIM) by 1.3% and perceptual quality (LPIPS) by 21% without incurring any additional computational overhead or requiring further fine-tuning.

Index Terms—Video Generation, Diffusion Models, Flow Matching, Classifier-Free Guidance, Zero-Shot Learning

I. INTRODUCTION

Text-to-video (T2V) generation has seen rapid progress following the success of text-to-image (T2I) diffusion models. The transition from U-Net architectures to Diffusion Transformers (DiTs) has enabled the scaling of models to billions of parameters, yielding high-fidelity video outputs. Open-weight models like Mochi-1-preview represent the state-of-the-art in accessible T2V generation.

Despite these architectural advances, open-weight video models frequently exhibit temporal instability (flickering), geometric distortion (morphing), and failure to adhere strictly to textual conditioning. Traditional solutions involve post-processing techniques such as deep-learning frame interpolation (e.g., RAFT) or spatial super-resolution. However, our preliminary investigations revealed that applying post-hoc interpolation to DiT-generated latent sequences often destroys the underlying video structure, severely degrading Structural Similarity (SSIM).

To address this, we propose a purely native, inference-time enhancement pipeline. Our approach requires no additional training, fine-tuning, or external post-processing networks. Instead, we manipulate the inherent mechanisms of the diffusion process:

- 1) **Scheduler Optimization:** We subtly modify the timestep weighting of the Flow Match Euler Discrete Scheduler to prioritize high-frequency detail resolution.
- 2) **Asymmetric Negative Prompting:** We inject structured negative embeddings to actively guide the attention blocks away from artifact-heavy latent subspaces.
- 3) **Guidance Amplification:** We increase the classifier-free guidance (CFG) scale to enforce stronger semantic alignment.

We demonstrate that these zero-cost modifications collectively produce a significant improvement in both temporal consistency and prompt fidelity.

II. RELATED WORK

A. Diffusion-Based Video Generation

Diffusion models operate by iteratively denoising a normally distributed variable to match a target data distribution. Video diffusion extends this by incorporating temporal attention mechanisms. Models such as Make-A-Video, Imagen Video, and Sora have demonstrated the potential of this paradigm. Mochi-1 represents a shift towards open-weight Asymmetric Diffusion Transformers (AsymmDiT), utilizing Flow Matching rather than traditional score-based formulations.

B. Inference-Time Guidance

Classifier-Free Guidance (CFG) is a standard technique to improve the alignment between generated samples and conditioning text by extrapolating in the direction of the conditional estimate away from an unconditional estimate. Negative prompting extends CFG by replacing the unconditional estimate with an explicitly undesirable semantic concept. While common in T2I, its systematic application in T2V to suppress specific temporal artifacts (e.g., "morphing", "jittery") remains under-explored.

III. METHODOLOGY

Our enhancements are applied directly within the reverse diffusion loop of the Mochi-1 pipeline.

A. Scheduler Optimization (Shift Adjustment)

Mochi-1 utilizes a Flow Match Euler Discrete Scheduler. The ‘shift’ parameter dictates the distribution of timesteps sampled during inference. By default, Mochi utilizes a shift of 1.0. We increase this parameter to its maximum intended value of 1.15. This subtle adjustment alters the timestep weighting, forcing the model to spend proportionally more capacity resolving structural geometry and high-frequency details before settling the temporal motion.

B. Structured Negative Prompt Injection

Early video models frequently struggle with object permanence, leading to “melting” or morphing artifacts. We inject a highly structured negative prompt: *“blurry, morphed, morphing, jittery, low resolution, deformed, unnatural motion, artifacts, bad anatomy”*. During CFG calculation, the text encoder generates embeddings for both the positive prompt and this negative prompt. The diffusion trajectory is actively pushed away from these artifact-associated regions of the latent space.

C. Guidance Amplification

To complement the negative prompt, we increase the CFG scale from the default 4.5 to 6.0. This stronger guidance enforces stricter adherence to the positive semantic prompt while penalizing deviations toward the negative embeddings.

IV. EXPERIMENTAL SETUP

A. Evaluation Prompts

To ensure robustness, we evaluate our pipeline on five diverse prompts encompassing various subjects, lighting conditions, and motion dynamics:

- 1) *“A camel moving in the desert with pyramids in the background, dramatic sunset”*
- 2) *“A golden retriever running through a field of wildflowers, sunny day”*
- 3) *“Ocean waves crashing on a rocky coastline during a storm, dramatic lighting”*
- 4) *“A butterfly landing on a flower in a garden, macro shot, soft focus background”*
- 5) *“A busy city street at night with neon signs and rain reflections”*

B. Metrics

We utilize four primary metrics:

- **CLIP-SIM:** Measures the cosine similarity between the generated frames and the textual prompt using the CLIP Vision/Text encoders. Evaluates semantic alignment.
- **SSIM (Structural Similarity Index):** Calculated between consecutive frames to measure structural preservation and identify flickering.
- **LPIPS (Learned Perceptual Image Patch Similarity):** Calculated between consecutive frames using a VGG network to quantify perceptual temporal consistency.
- **FVD (Fréchet Video Distance):** Measures the distributional distance between the baseline and enhanced videos using InceptionV3 features.

C. Ablation Design and Hardware

We isolate each enhancement in a comprehensive ablation study:

- **Baseline:** Default configuration (Shift 1.0, CFG 4.5, No Neg Prompt).
- **Config A:** Scheduler shift only (1.15).
- **Config B:** Negative prompt only.
- **Config C:** Guidance scale only (6.0).
- **Config D:** All enhancements combined.

Each of the 5 configurations is run across all 5 prompts using 3 distinct random seeds (42, 123, 456), yielding 75 total video generations. All experiments generate 60 frames at 30 fps using ‘bfloat16’ precision. Experiments were executed in parallel on a cluster of NVIDIA RTX A6000 (48GB) GPUs. To maximize inference speed, CPU offloading was strategically managed to retain the Transformer in VRAM during denoising while offloading during the final VAE decode, utilizing ‘expandable_segments’ to mitigate memory fragmentation.

V. RESULTS

A. Quantitative Analysis

Our results (Table ??) demonstrate the efficacy of our zero-cost enhancements.

The combined configuration (Config D) achieved the highest temporal stability, realizing a significant reduction in LPIPS (0.0410 vs 0.0522 baseline, a 21% improvement) and the highest average SSIM (0.9511). This confirms that the combination of scheduler shifting and negative prompting effectively suppresses the high-frequency jitter and morphing artifacts characteristic of the baseline model.

Regarding semantic alignment, Config C (Guidance Only) achieved the highest overall CLIP-SIM (33.13), which is expected given the higher CFG scale. The combined approach (Config D) maintained a highly competitive CLIP-SIM (32.65) while drastically outperforming Config C in temporal metrics (LPIPS 0.0410 vs 0.0575).

Importantly, our enhancements introduced zero computational overhead. Average inference time for the combined configuration was slightly lower than the baseline (476.3s vs 492.4s), primarily due to optimizations in VRAM allocation.

TABLE I
ABLATION STUDY RESULTS (MEAN $\pm$ STD). BEST RESULTS IN BOLD.

Config	CLIP-SIM	SSIM	LPIPS	Time (s)
Baseline	32.09 $\pm$ 2.10	0.9393 $\pm$ 0.0715	0.0522 $\pm$ 0.0514	492.4
A: Scheduler Only	31.98 $\pm$ 2.00	0.9487 $\pm$ 0.0462	0.0462 $\pm$ 0.0421	488.5
B: Neg Prompt Only	32.67 $\pm$ 2.47	0.9509 $\pm$ 0.0498	0.0439 $\pm$ 0.0412	481.8
C: Guidance Only	33.13 $\pm$ 2.37	0.9404 $\pm$ 0.0679	0.0575 $\pm$ 0.0699	476.0
D: All Combined	32.65 $\pm$ 2.70	0.9511 $\pm$ 0.0516	0.0410 $\pm$ 0.0413	476.3

B. Visual and Distributional Impact

We analyzed the Fréchet Video Distance (FVD) to measure how our enhancements shifted the generated video distribution relative to the baseline. Config A (Scheduler Only) yielded the

lowest FVD (90.87), indicating that the scheduler shift preserved the core motion dynamics of the baseline while refining details. Configs involving guidance shifts or negative prompts (B, C, D) resulted in larger FVDs (136-150), demonstrating a deliberate divergence from the baseline distribution toward higher-fidelity spaces free of semantic artifacts.

Visual inspection confirms these quantitative findings. (Reference Figure ?? in the full repository). Baseline videos frequently exhibited "melting" textures on background elements, whereas Config D consistently maintained rigid geometry and sharp feature boundaries across the full 60-frame sequence.

C. Discussion

The ablation study highlights a critical trade-off in video diffusion models. Increasing CFG (Config C) strongly improves prompt adherence but significantly degrades temporal consistency (worst LPIPS: 0.0575). However, by coupling high CFG with targeted negative prompting and scheduler optimization (Config D), we successfully mitigate this degradation, achieving best-in-class temporal stability without sacrificing the semantic gains.

VI. CONCLUSION

We present a zero-cost, training-free enhancement methodology for open-weight video diffusion models. By applying independent, synergistic modifications to the inference scheduler, text conditioning, and guidance scales, we significantly improve the temporal consistency and structural integrity of generated videos. Our rigorous ablation study confirms that these native edits outperform the baseline model across both perceptual and structural metrics without any additional computational burden.